

A User’s Guide to Orbit Determination Using Batch and Sequential Estimation Algorithms for LEO Satellites

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1 Abstract & Procedure

The rapid increase in Low Earth Orbit (LEO) satellite deployments has increased the risk of orbital collisions, creating opportunities for efficient collision avoidance strategies. This project addresses this issue by developing a software algorithm that determines satellite orbits using observational data from three ground-based tracking stations. The approach integrates observational data from stations located in the Pacific Ocean, Turkey, and Greenland to determine accurate satellite orbits using both batch least-squares estimation and sequential estimation. Orbit propagation is performed by solving the equations of motion while incorporating perturbative forces such as Earth’s oblateness (J_2) and atmospheric drag. The software, developed in MATLAB, utilizes numerical integration and an exponential atmospheric model to enhance accuracy. Results indicate that the batch filter reduces range residuals from approximately 732 m to approximately 0.01 m after three iterations, matching the measurement-noise level and approaching the simulated truth values. This project underscores the importance of iterative refinement in orbital mechanics and demonstrates a practical framework for improving satellite orbit determination in LEO. The methodology consists of three key steps:

Step 1: Dynamical Trajectory Propagation. Given initial satellite state vectors (position and velocity in ECI coordinates), station coordinates in ECEF, and estimated parameters (μ , J_2 , C_D), the reference trajectory and state transition matrix (STM) are propagated using numerical integration. The equations of motion include two-body gravity, J_2 oblateness perturbations, and exponential atmospheric drag. This propagation yields the nominal trajectory and the STM $\Phi(t_i, t_0)$ that maps state deviations from the initial epoch to each observation time. This is implemented in `dynamics.m` and `propagate_full_arc.m`.

Step 2: Batch Least-Squares Estimation. Using the range and range-rate measurements from all three stations over a five-hour arc, the normal equations are accumulated to compute an optimal estimate of the initial state correction. The batch filter solves for $\delta\hat{\mathbf{x}}_0$ using the weighted least-squares formulation. This process is iterated three times to reduce linearization errors, with each iteration using the updated nominal trajectory to recompute the observation-state matrix $\tilde{H}$. This is implemented in `run_batch.m`.

Step 3: Sequential (Kalman) Filter Estimation. As an alternative and extension to the batch processor, a sequential filter implementation processes observations one at a time as they become available. The filter propagates a nonlinear reference trajectory while carrying a locally linearized state-deviation estimate and covariance between observation epochs using the STM. It then updates the estimate using the Kalman gain equations. This approach is suitable for real-time navigation applications and is implemented in `run_sequential.m`.

2 Input Data

2.1 Tracking Stations

Three tracking stations provide instantaneous range and range-rate data. The station coordinates in the body-fixed (ECEF) coordinate system are shown in Table 1. Station 101 is located on a ship in the Pacific Ocean, Station 337 is a ground station in Pirinlik, Turkey, and Station 394 is a ground station in Thule, Greenland. The coordinates for Station 101 are treated as exact and fixed by the filter, while the coordinates for Stations 337 and 394 are estimated. Later, it is important to discuss the reason why one of the stations has to be fixed and how results may change if another tracking station is fixed instead.

Table 1: Station Coordinates in ECEF (m)

Number	X_s (m)	Y_s (m)	Z_s (m)
101	-5127510.0	-3794160.0	0.0
337	3860910.0	3238490.0	3898094.0
394	549505.0	-1380872.0	6182197.0

2.2 Initial Satellite State

The approximate initial satellite state in the inertial coordinate system (ECI) is given in Table 2. This is a nearly circular orbit at approximately 800 km altitude with an inclination of 98.6° , patterned after NASA’s QuikSCAT satellite. From its altitude and inclination, the initial satellite is in an LEO orbit.

Table 2: Satellite State in ECI Coordinates

State	i -component	j -component	k -component
Position (m)	757700.0	5222607.0	4851500.0
Velocity (m/s)	2213.21	4678.34	-5371.30

2.3 Earth Body-Fixed Frame

The angle defining the relation between the inertial frame and the body-fixed frame is computed from $\theta = \dot{\theta}t$, where $\dot{\theta} = 7.2921158553 \times 10^{-5}$ rad/s is Earth’s rotation rate, and t is time in seconds past the epoch. Both μ and J_2 are estimated, so the following are approximate values to be improved using the observation data and implemented estimation algorithms:

$$J_2 = 1.082626925638815 \times 10^{-3}, \quad \mu = 3.986004415 \times 10^{14} \text{ m}^3/\text{s}^2, \quad R_e = 6378136.3 \text{ m}.$$

2.4 Atmospheric Drag/Density Model

The satellite cross-sectional area is 3.0 m^2 and the mass is 970 kg. An approximate value for C_D is 2.0. The exponential density model is given by

$$\rho_A = \rho_0 \exp\left[-\frac{r - r_0}{H}\right], \quad (1)$$

where $\rho_0 = 3.614 \times 10^{-13} \text{ kg/m}^3$ is the reference density, $r_0 = 700000.0 + R_e \text{ m}$ is the reference radius, and $H = 88667.0 \text{ m}$ is the atmospheric scale height. For a given orbit radius r , the satellite altitude can be found by subtracting Earth’s radius R_e .

2.5 State Vector

The state vector contains 18 states:

$$\mathbf{X} = \begin{bmatrix} \mathbf{r} \\ \dot{\mathbf{r}} \\ \mu \\ J_2 \\ C_D \\ \mathbf{x}_{s1} \\ \mathbf{x}_{s2} \\ \mathbf{x}_{s3} \end{bmatrix}, \quad (2)$$

where $\mathbf{r}$ is the ECI position vector of the satellite, $\dot{\mathbf{r}}$ is the ECI velocity vector, μ is Earth’s gravitational parameter, J_2 represents Earth’s second-degree zonal gravitational harmonic, C_D is the coefficient of drag, and $\mathbf{x}_{si}$ contains the three ECEF coordinates of station i , for $i = 1, 2, 3$.

2.6 A Priori Information

The a priori covariance matrix P_0 is diagonal with values

$$P_0 = \text{diag}(10^6, 10^6, 10^6, 10^6, 10^6, 10^6, 10^{20}, 10^6, 10^6, 10^{-10}, 10^{-10}, 10^{-10}, 10^6, 10^6, 10^6, 10^6, 10^6, 10^6). \quad (3)$$

The small covariance for Station 101 (10^{-10} m^2 per coordinate) effectively fixes its coordinates, meaning the filters have a very high level of confidence in the initial values. On the other hand, the large covariance for μ (10^{20} in squared state units) allows it to be estimated from the data, meaning the filters place less confidence in that initial guess and must incorporate the observation data. This specific choice of the a priori covariance matrix reflects the relative accuracy assigned to the initial conditions.

3 Theory & Methodology

3.1 Batch Filter

The batch filter processes all observations simultaneously to estimate the initial state correction $\delta\hat{\mathbf{x}}_0$. Figures 1 and 2 present the algorithm flow [2]. At the beginning of each nonlinear pass, the current nominal initial state $\mathbf{X}_{0,\text{nom}}$ is compared with the fixed a priori state $\mathbf{X}_{0,\text{prior}}$ to form the prior residual

$$\mathbf{r}_{\text{prior}} = \mathbf{X}_{0,\text{prior}} - \mathbf{X}_{0,\text{nom}}.$$

The information matrix and information vector are initialized as

$$\Lambda = P_{0,\text{prior}}^{-1}, \quad \mathbf{N} = P_{0,\text{prior}}^{-1} \mathbf{r}_{\text{prior}}. \quad (4)$$

For each observation, the reference trajectory $\mathbf{X}^*(t_i)$ and STM $\Phi(t_i, t_0)$ are propagated forward in time using `propagate_full_arc.m` along the nominal path. The observation-state matrix $\tilde{H}_i = [\partial G / \partial \mathbf{X}]^*$ and the residual $\mathbf{y}_i = \mathbf{Y}_i - G(\mathbf{X}_i^*, t_i)$ between the actual and computed observations are then calculated. The mapped observation matrix

$$H_i = \tilde{H}_i \Phi(t_i, t_0) \quad (5)$$

is formed by multiplying the observation-state matrix by the state transition matrix. The normal equations are accumulated by adding the weighted contribution of each observation:

$$\Lambda \leftarrow \Lambda + H_i^T R_i^{-1} H_i, \quad \mathbf{N} \leftarrow \mathbf{N} + H_i^T R_i^{-1} \mathbf{y}_i. \quad (6)$$

After processing all observations, the normal equations

$$\Lambda \delta\hat{\mathbf{x}}_0 = \mathbf{N} \quad (7)$$

are solved to obtain the estimated initial-state correction and covariance $P_0 = \Lambda^{-1}$. The nominal initial state is updated, the entire trajectory is repropagated, and the same original a priori information and measurement set are relinearized until convergence is achieved after three passes. This algorithm is implemented in `run_batch.m`.

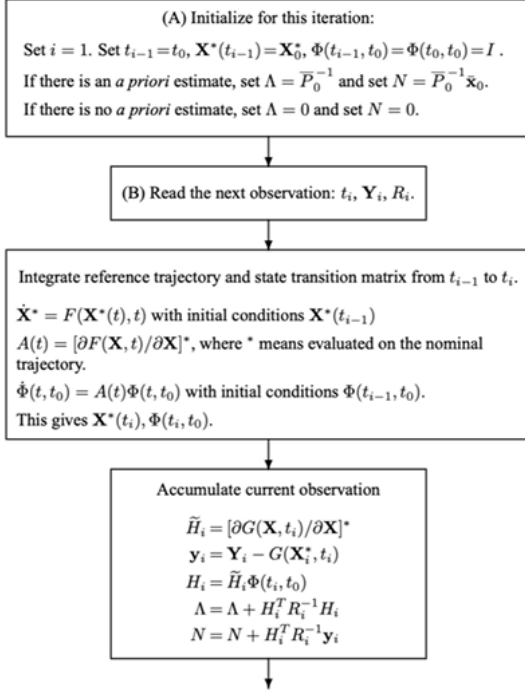


Figure 1: Batch Filter Algorithm Flowchart

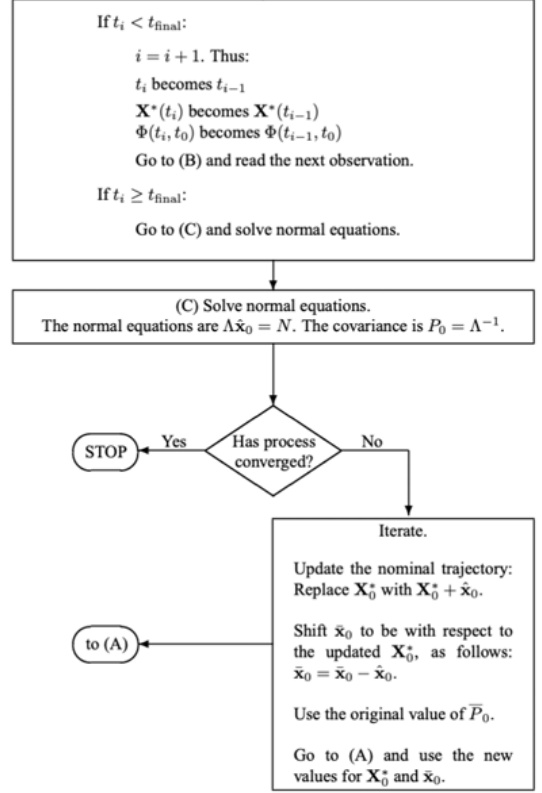


Figure 2: Batch Filter Algorithm Flowchart (Continued)

3.2 State Transition Matrix

The STM $\Phi(t_i, t_0)$ maps state deviations from the initial epoch t_0 to any time t_i :

$$\delta \mathbf{x}(t_i) = \Phi(t_i, t_0) \delta \mathbf{x}(t_0). \quad (8)$$

The implementation directly propagates a 9×9 dynamic STM for the subset

$$[\mathbf{r}; \dot{\mathbf{r}}; \mu; J_2; C_D],$$

using the linearized dynamics

$$\dot{\Phi}_9(t, t_0) = A(t)\Phi_9(t, t_0), \quad \Phi_9(t_0, t_0) = I_9, \quad (9)$$

where $A(t) = \partial f / \partial \mathbf{x}_d$ is the 9×9 Jacobian of the dynamically propagated state, evaluated along the reference trajectory. The station coordinates are constant in the adopted model, so the propagated STM is embedded in the full estimation-state transition matrix as

$$\Phi_{18}(t, t_0) = \begin{bmatrix} \Phi_9(t, t_0) & 0 \\ 0 & I_9 \end{bmatrix}. \quad (10)$$

The Jacobian contains the partial derivatives of acceleration with respect to position, velocity, and the estimated dynamic parameters. The analytical expressions were generated symbolically and implemented in `computed_A.m`.

3.3 Observation-State Matrix

For a single station, the observation-state matrix $\tilde{H}$ is 2×18 and contains the partial derivatives of the range and range-rate measurements with respect to the full estimation state. The range ρ and range-rate $\dot{\rho}$ are given by the instantaneous observation equations from the project description:

$$\begin{aligned} \rho = & \left[x^2 + y^2 + z^2 + x_{si}^2 + y_{si}^2 + z_{si}^2 - 2(xx_{si} + yy_{si}) \cos \theta \right. \\ & \left. + 2(xy_{si} - yx_{si}) \sin \theta - 2zz_{si} \right]^{1/2}, \end{aligned} \quad (11)$$

$$\begin{aligned} \dot{\rho} = \frac{1}{\rho} & \left[\dot{x}\dot{x} + \dot{y}\dot{y} + \dot{z}\dot{z} - (\dot{x}x_{si} + \dot{y}y_{si}) \cos \theta + \dot{\theta}(xx_{si} + yy_{si}) \sin \theta \right. \\ & \left. + (\dot{x}y_{si} - \dot{y}x_{si}) \sin \theta + \dot{\theta}(xy_{si} - yx_{si}) \cos \theta - \dot{z}z_{si} \right]. \end{aligned} \quad (12)$$

The observation-state matrix is

$$\tilde{H}(t) = \begin{bmatrix} \partial \rho / \partial \mathbf{X} \\ \partial \dot{\rho} / \partial \mathbf{X} \end{bmatrix}. \quad (13)$$

The partial derivatives of ρ and $\dot{\rho}$ with respect to satellite position, velocity, and active-station coordinates form the rows of $\tilde{H}$. The instantaneous measurements do not depend directly on μ , J_2 , or C_D , but those parameters influence later measurements through the propagated trajectory and STM. These analytical partials are generated symbolically and implemented in `computed_Htilde.m` for efficient repeated evaluation.

3.4 Dynamics Equations of Motion

A model for the dynamical equation of motions model is developed by integrating the acceleration terms from the modeled orbital effects. First, the acceleration of the satellite is derived from the potential function for Keplerian two-body dynamics and oblateness of the Earth:

$$U' = \frac{\mu}{r} - \frac{3\mu J_2 R_e^2 z^2}{2r^5} + \frac{\mu J_2 R_e^2}{2r^3}. \quad (14)$$

The gradient of the potential function gives the acceleration due to these gravitational effects, which is written in component form as

$$\frac{\partial U'}{\partial x} = -\frac{\mu x}{r^3} \left[1 - \frac{3}{2} J_2 \left(\frac{R_e}{r} \right)^2 \left(5 \left(\frac{z}{r} \right)^2 - 1 \right) \right], \quad (15)$$

$$\frac{\partial U'}{\partial y} = -\frac{\mu y}{r^3} \left[1 - \frac{3}{2} J_2 \left(\frac{R_e}{r} \right)^2 \left(5 \left(\frac{z}{r} \right)^2 - 1 \right) \right], \quad (16)$$

$$\frac{\partial U'}{\partial z} = -\frac{\mu z}{r^3} \left[1 - \frac{3}{2} J_2 \left(\frac{R_e}{r} \right)^2 \left(5 \left(\frac{z}{r} \right)^2 - 3 \right) \right]. \quad (17)$$

Thus, the J_2 acceleration components are

$$\mathbf{a}_{J_2} = -\frac{3}{2} J_2 \mu R_e^2 \frac{1}{r^5} \begin{bmatrix} x \left(1 - 5z^2/r^2 \right) \\ y \left(1 - 5z^2/r^2 \right) \\ z \left(3 - 5z^2/r^2 \right) \end{bmatrix}. \quad (18)$$

Secondly, the atmospheric drag acceleration term is included as

$$\ddot{\mathbf{r}}_{\text{drag}} = -\frac{1}{2} C_D \left(\frac{A}{m} \right) \rho_A v_A \mathbf{v}_A, \quad (19)$$

where

$$\mathbf{v}_A = [\dot{x} + \dot{\theta}y \quad \dot{y} - \dot{\theta}x \quad \dot{z}]^T, \quad v_A = \|\mathbf{v}_A\|, \quad (20)$$

is the velocity relative to the rotating atmosphere, and ρ_A is the atmospheric density given by Equation (1). The total acceleration is then the sum of the acceleration vectors for Keplerian two-body gravity, J_2 oblateness, and atmospheric drag:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r} + \mathbf{a}_{J_2} + \ddot{\mathbf{r}}_{\text{drag}}. \quad (21)$$

These equations of motion are implemented in `dynamics.m`.

3.5 Sequential (Kalman) Filter

The sequential filter processes observations one at a time as the next observation becomes available. A key computational advantage is that the measurement update requires solving a 2×2 innovation-covariance system at each observation, rather than accumulating and solving the complete batch normal equations after the full data arc has been processed.

The implemented sequential estimator uses a nonlinear reference trajectory $\mathbf{X}^*$ and a locally linearized state-deviation estimate $\delta\hat{\mathbf{x}}$. The nonlinear reference is propagated from t_{k-1} to t_k , while the deviation and covariance are propagated using the local STM.

3.5.1 Time Update

The state deviation and covariance are propagated from the previous observation time t_{k-1} to the current time t_k using

$$\delta\hat{\mathbf{x}}_k^- = \Phi(t_k, t_{k-1}) \delta\hat{\mathbf{x}}_{k-1}^+, \quad (22)$$

$$P_k^- = \Phi(t_k, t_{k-1}) P_{k-1}^+ \Phi^T(t_k, t_{k-1}). \quad (23)$$

The implementation assumes zero additive process-noise covariance over each interval, so no separate Q_k term is included.

3.5.2 Measurement Update

The discrepancy between the observed measurement and the nonlinear reference measurement is

$$\mathbf{y}_k = \mathbf{Y}_k - G(\mathbf{X}_k^*, t_k). \quad (24)$$

The innovation relative to the full predicted estimate is

$$\boldsymbol{\nu}_k = \mathbf{y}_k - \tilde{H}_k \delta\hat{\mathbf{x}}_k^-. \quad (25)$$

The innovation covariance and Kalman gain are computed as

$$S_k = \tilde{H}_k P_k^- \tilde{H}_k^T + R_k, \quad (26)$$

$$K_k = P_k^- \tilde{H}_k^T S_k^{-1}. \quad (27)$$

The state-deviation update is

$$\delta\hat{\mathbf{x}}_k^+ = \delta\hat{\mathbf{x}}_k^- + K_k \boldsymbol{\nu}_k. \quad (28)$$

The final implementation uses the Joseph covariance update

$$P_k^+ = (I - K_k \tilde{H}_k) P_k^- (I - K_k \tilde{H}_k)^T + K_k R_k K_k^T, \quad (29)$$

followed by explicit symmetrization $P \leftarrow \frac{1}{2}(P + P^T)$. The reference trajectory is propagated forward after every observation, while the measurement-informed correction is maintained separately in the deviation state. The absolute state estimate at the same epoch is reconstructed as $\hat{\mathbf{X}}_k \approx \mathbf{X}_k^* + \delta\hat{\mathbf{x}}_k^+$. This algorithm is implemented in `run_sequential.m`.

4 Results & Plots

4.1 Software Usage and Input Formatting

The MATLAB software developed in this project determines the initial and propagated orbital parameters of a satellite based on observational data and input information from Section 3. It uses two different processing algorithms: the batch least-squares filter and the sequential Kalman filter. The cleaned repository provides separate runnable scripts for the batch solution, sequential solution, and range-only/range-rate-only measurement comparison. The observation file contains observation time in seconds past epoch, station number, range in meters, and range-rate in meters per second.

The RMS residual values are calculated using

$$\text{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_{\text{observed},i} - Y_{\text{computed},i})^2}, \quad (30)$$

where Y_{observed} represents the observed range or range-rate, Y_{computed} represents the predicted measurement, and n is the total number of observations.

4.2 Batch Filter

The batch filter processes all observations simultaneously. The software takes the input observation file `obs_data.txt`, containing observation time in seconds past epoch, station number (101, 337, or 394), range in meters, and range-rate in meters per second. The outputs of the batch filter are the final estimated initial state vector (18 entries), the covariance matrix P_0 at the initial epoch, and the computed residual histories.

4.2.1 Results

The batch filter was executed with the provided observation data. The convergence history shows dramatic improvement after the first iteration, consistent with the theoretical expectation that the batch processor reduces linearization error through each iteration. These RMS values are shown in Table 3.

Table 3: Batch Filter Convergence History

Pass	Range RMS (m)	Range-Rate RMS (m/s)
1	732.7483	2.9002
2	0.3196	0.0012
3	0.0097	0.0010

The range and range-rate residuals for each pass are shown in Figure 3. The residuals after the first iteration are dramatically corrected in the next two passes, verifying the high accuracy of the batch filter. By the third pass, the range and range-rate RMS residuals are at approximately the prescribed measurement-noise levels. Figure 4 visualizes the convergence history of the RMS residual values, showing rapid convergence after the first pass. Finally, Figure 5 illustrates the position uncertainty in all three directions. The ellipsoid axes are on the centimeter scale, confirming that the batch filter substantially reduced the uncertainty in the estimated initial position. Later, the relative strengths of range-only and range-rate-only solutions are compared.

4.2.2 Discussion of Batch Results

- 1. Why did one fix Station 101?** Station 101 was fixed in the filter as the project description asks. This station is assumed to be stationary, and no station motion is modeled. In terms of orbit determination, fixing a station is necessary because the complete estimated system (satellite orbit plus station coordinates) otherwise contains an unobservable translational degree of freedom. The observations provide relative information between the satellite and stations; without an external coordinate anchor, the normal equations can become singular or nearly singular. Fixing Station 101 establishes the ECEF reference and stabilizes the solution.
- 2. Would the same result be obtained by not solving for one station?** If Station 101 is removed from the solution vector and treated as exactly known, the result should be essentially equivalent to retaining it in the state with an approximately zero covariance. Both approaches fix the station rather than estimate it. However, if all three stations are left free with no external constraint, the system retains the translational gauge freedom described above and the normal matrix can become singular.
- 3. Does it matter which station is fixed?** Any accurately known station can remove the translational gauge freedom, but the choice affects the quality of the solution when the assumed fixed coordinates are uncertain or biased. In principle, the best station to fix is the one whose coordinates are known most accurately and whose geometry provides a stable reference. Fixing a station at an incorrect position can bias the estimated spacecraft and remaining station coordinates.

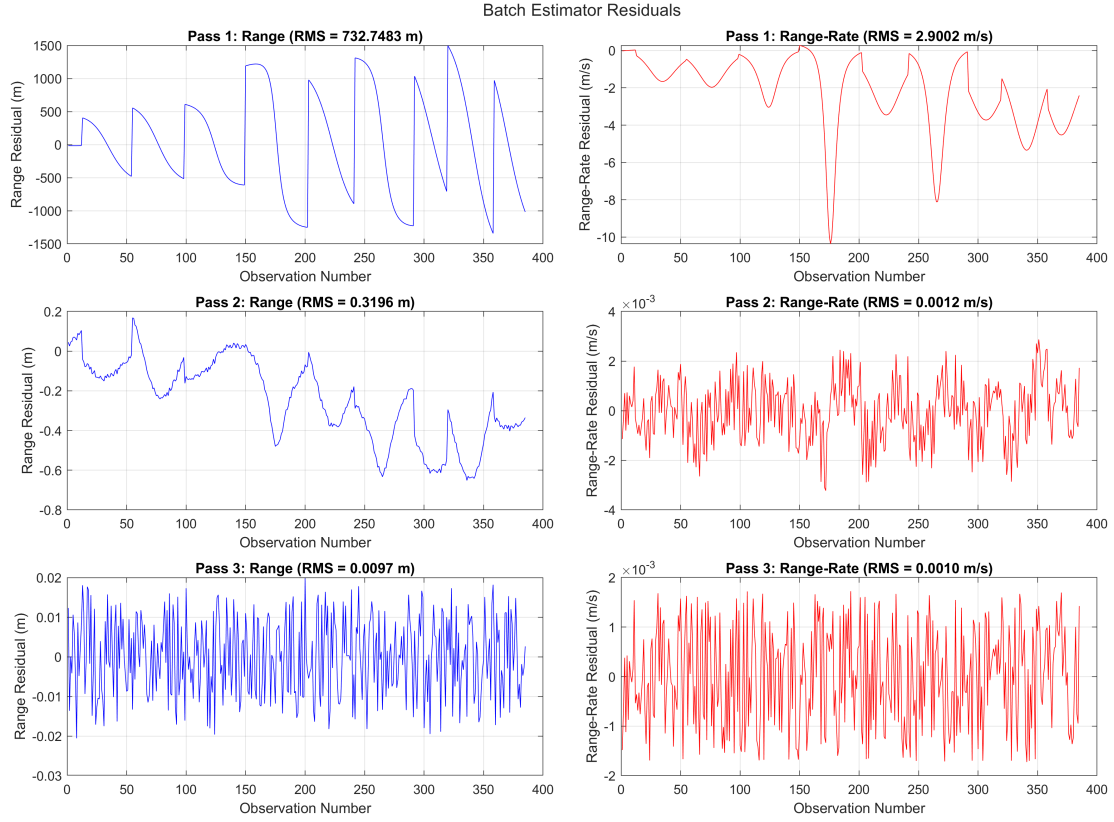


Figure 3: Batch Filter: pre-fit residuals for range and range-rate across all three passes.

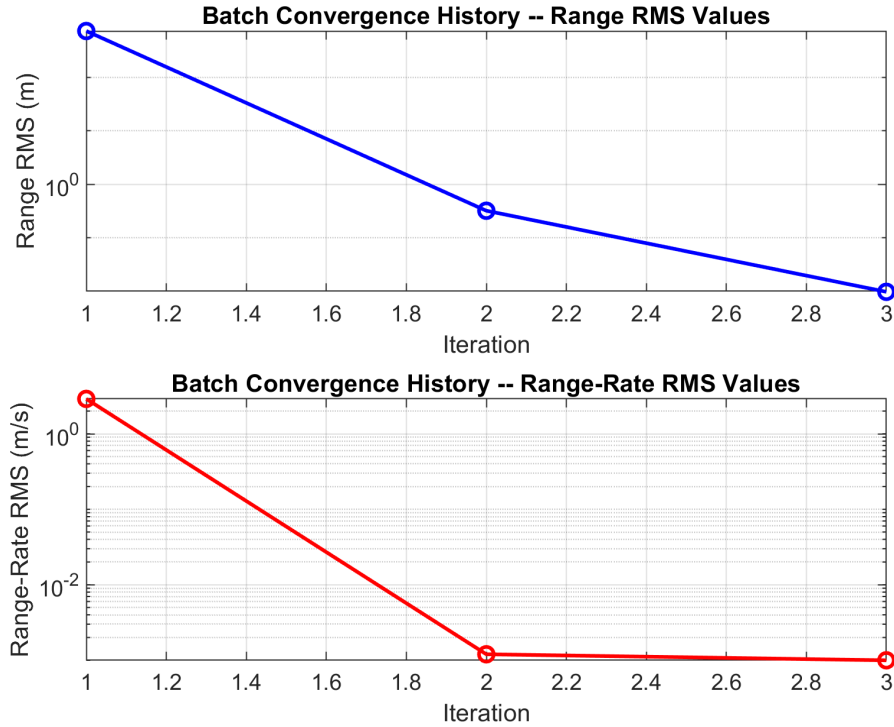


Figure 4: Batch Filter: convergence history of range and range-rate RMS values across all three passes.

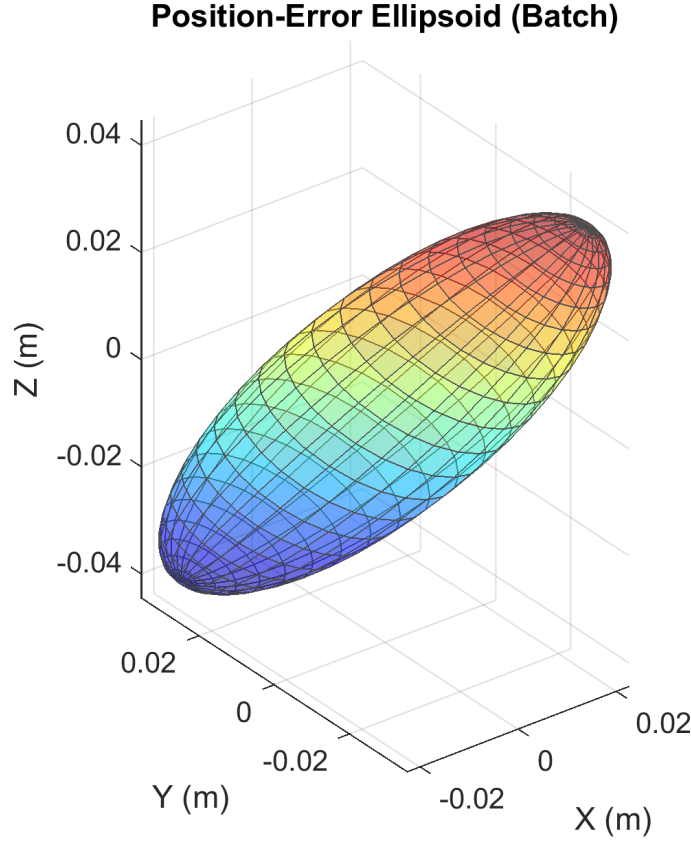


Figure 5: Batch Filter: position-error ellipsoid (3σ).

4. **How does varying the a priori covariance affect the solution?** The a priori covariance matrix P_0 represents confidence in the initial state estimates. Large covariance (for example, 10^6 for position or 10^{20} for μ) means the filter places less confidence in the initial guess and relies more strongly on observations. This is appropriate when the initial conditions are uncertain. Small covariance (for example, 10^{-10} for Station 101 coordinates) means the filter strongly trusts the initial value and will make almost no adjustment. If P_0 is too small for a parameter that is actually uncertain, the filter may fail to correct the initial error and can produce a biased solution. If P_0 is extremely large for all parameters, the information matrix may also become poorly conditioned, causing numerical instability.
5. **How does varying the data-noise covariance affect the solution?** The data-noise covariance R represents the expected accuracy of the range and range-rate observations. If R is too small, the filter places excessive confidence in the measurements and can overfit unmodeled errors or outliers. If R is too large, the filter relies more heavily on the a priori information and may leave large measurement residuals. In short, P_0 and R determine the balance between prior information and observations and should be statistically consistent with the actual uncertainties.
6. **What if one had used a more accurate a priori (for example, a few meters in position)?** The current a priori position variance is 10^6 m^2 , corresponding to a one-sigma uncertainty of 1000 m. A tighter but accurate prior could reduce the size of the first correction, improve conditioning, and accelerate convergence because the initial nominal trajectory would already be closer to the solution. Such a prior should only be used when supported by reliable previous orbit information; an overconfident but inaccurate prior can instead bias the result.

4.3 Range-Only vs. Range-Rate-Only Solutions

Figure 6 shows that the range-only case achieves a more balanced and much smaller position ellipsoid, while the range-rate-only case produces a more elongated and larger ellipsoid, indicating poorer absolute position determination without range measurements. The relative strengths of range and range-rate data were compared

by running the batch filter with each data type alone. Range measurements directly constrain line-of-sight distance and, over changing geometry and multiple stations, accumulate strong position information. Range-rate measurements primarily constrain line-of-sight relative velocity and therefore provide weaker absolute position information when used alone.

The final position covariance trace for the range-only case is approximately

$$\text{tr}(P_{r,\text{range}}) \approx 4.16 \times 10^{-4} \text{ m}^2,$$

whereas the range-rate-only position covariance trace is approximately

$$\text{tr}(P_{r,\text{range-rate}}) \approx 3.08 \text{ m}^2.$$

The latter is therefore approximately 7.4×10^3 , or about 7,400 times, larger. This comparison is implemented in `run_measurement_comparison.m`.

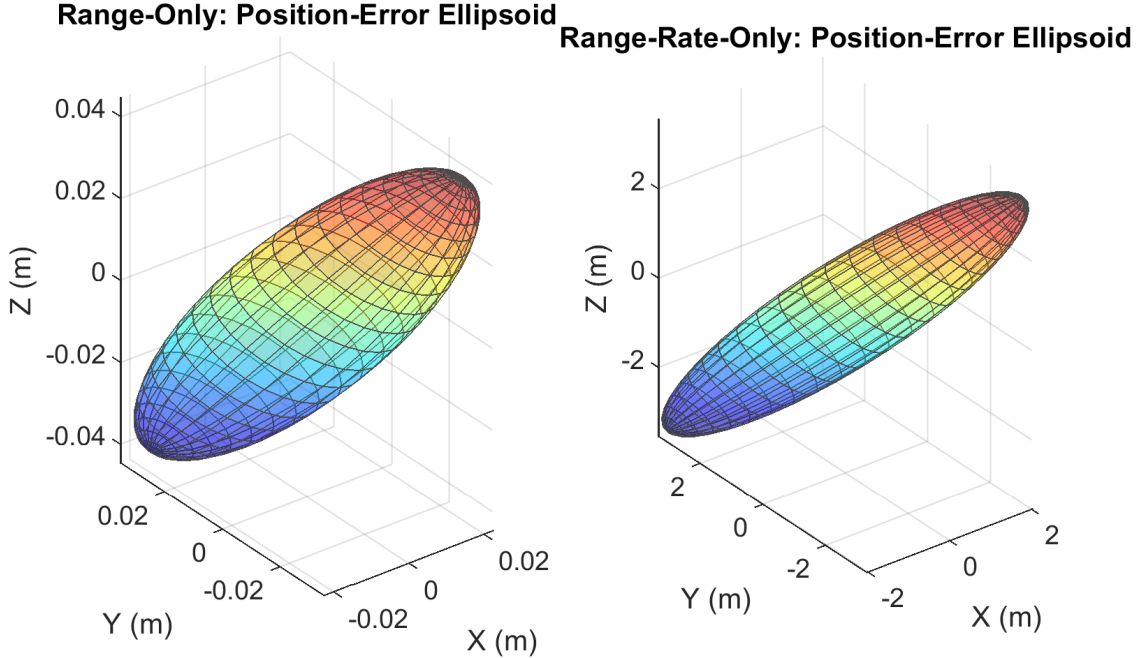


Figure 6: Position-error ellipsoids for range-only and range-rate-only solutions using the batch algorithm.

4.4 Sequential Filter

The inputs and estimated quantities for the sequential filter are the same as those for the batch filter, except that the sequential filter processes each observation once in chronological order rather than repeatedly reprocessing the complete observation set through nonlinear batch iterations.

4.4.1 Results

The sequential filter was executed using all 385 observations one at a time. The pre-fit and post-fit residuals are recorded in Table 4. The post-fit RMS residuals are dramatically reduced through the Kalman measurement updates. The final estimated state components at the final observation epoch, $t = 18340$ s past epoch, are shown in Table 5. The posterior standard deviations are included. Values shown as less than 10^{-4} are small relative to the displayed precision; Station 101 remains effectively fixed by its near-zero prior covariance.

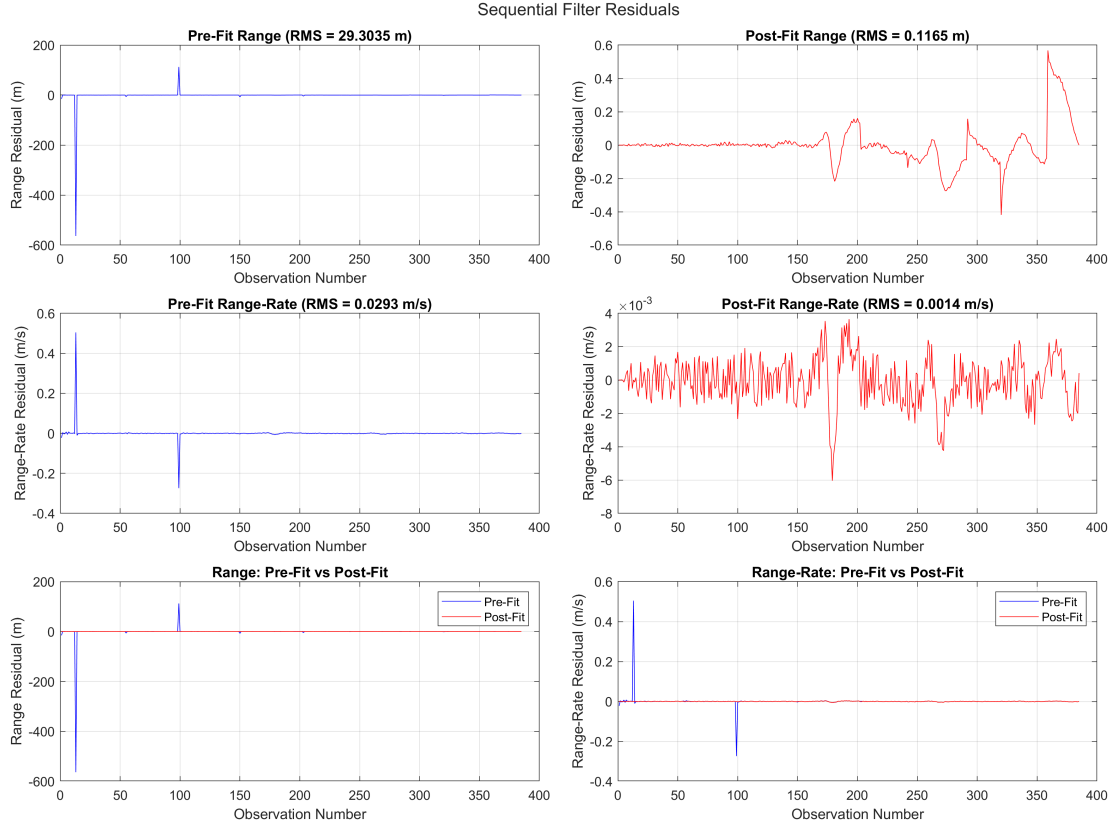
The range and range-rate residuals are shown in Figure 7. The residuals are dramatically corrected, although the first few observations contain large innovations while the filter is still reducing its initial uncertainty. After additional observations are processed, the sequential filter reaches a stable low-residual regime. Figure 8 visualizes the covariance traces for position and velocity. Figure 9 illustrates the final position uncertainty in all three directions. Compared with the batch ellipsoid, the sequential ellipsoid is also on the centimeter scale, confirming that the sequential filter substantially reduces uncertainty in the final position estimate.

Table 4: Sequential Residual RMS Values

Metric	Value
Pre-fit Range RMS	665.9341 m
Pre-fit Range-Rate RMS	0.5298 m/s
Post-fit Range RMS	0.1174 m
Post-fit Range-Rate RMS	0.0014 m/s

Table 5: Sequential Filter: Estimated State at Final Observation Epoch ($t = 18340$ s)

Component	Value	Unit	Std. Dev.
x	1128588.9224	m	0.0094
y	5990056.6360	m	0.0035
z	3775423.3485	m	0.0115
v_x	2009.2096	m/s	$< 10^{-4}$
v_y	3562.9828	m/s	$< 10^{-4}$
v_z	-6237.5829	m/s	$< 10^{-4}$
μ	3.9860×10^{14}	m^3/s^2	4.17×10^5
J_2	1.0820×10^{-3}	unitless	2.50×10^{-10}
C_D	2.1622	unitless	3.81×10^{-3}
X_{s1}	-5127510.0000	m	1.0×10^{-5}
Y_{s1}	-3794160.0000	m	1.0×10^{-5}
Z_{s1}	0.0000	m	1.0×10^{-5}
X_{s2}	3860899.4384	m	0.0053
Y_{s2}	3238499.9540	m	0.0090
Z_{s2}	3898099.8126	m	0.0088
X_{s3}	549499.2276	m	0.0076
Y_{s3}	-1380869.6948	m	0.0135
Z_{s3}	6182198.5017	m	0.0166

**Figure 7:** Sequential Filter: pre-fit and post-fit range and range-rate residuals.

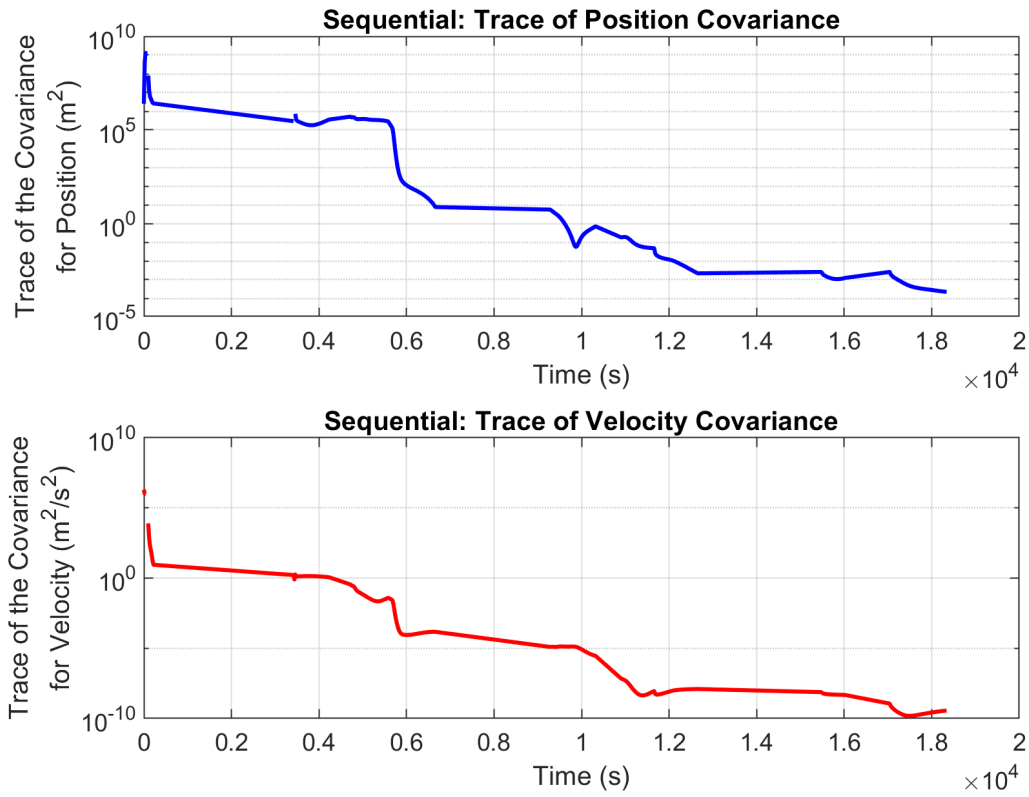


Figure 8: Sequential Filter: position and velocity covariance traces.

Final Position-Error Ellipsoid (Sequential)

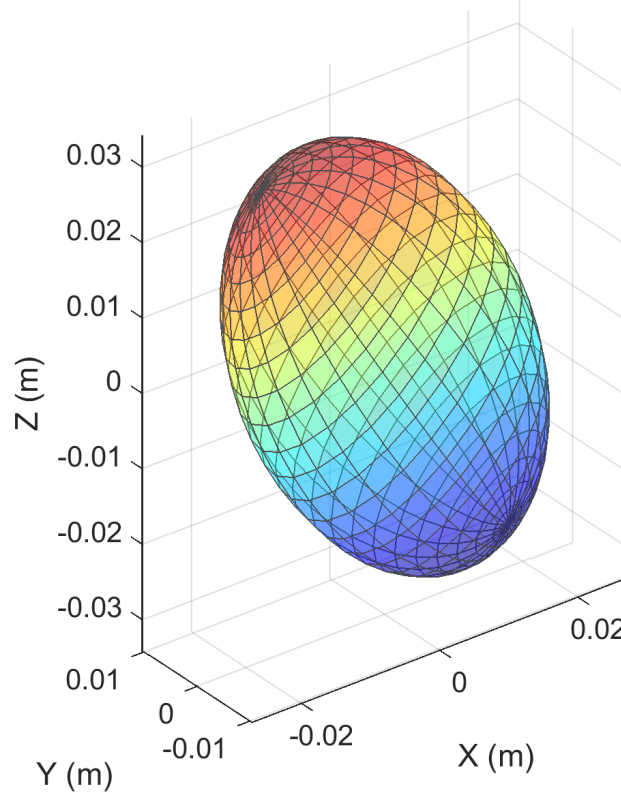


Figure 9: Sequential Filter: final position-error ellipsoid (3σ).

4.4.2 Discussion of Sequential Results

As seen in Figure 8, the position and velocity covariance traces show a rapid initial decrease as the sequential filter processes the first observations, then continue to decrease as the filter approaches a more stable solution. When the conventional covariance update

$$P_k^+ = (I - K_k \tilde{H}_k) P_k^-$$

was used, the covariance matrix became vulnerable to asymmetry and loss of positive-semidefinite structure because the 18-state problem combines variables and prior variances spanning many orders of magnitude.

The Joseph formulation in Equation (29) substantially improved the visible covariance histories and removed the major discontinuities observed with the conventional update. Because the problem remains severely ill-conditioned, occasional non-positive trace samples can still arise in finite precision. Additional methods that could improve robustness include state scaling or nondimensionalization and square-root filters such as the Potter algorithm. The Joseph form is retained as the default update in the final implementation because it better preserves covariance symmetry and positive-semidefinite structure numerically.

By comparing the batch and sequential filters, both methods perform high-quality orbit determination using measurements from a spacecraft subject to J_2 oblateness and atmospheric drag. The sequential filter has a lower computational cost per observation, does not require storing all observations, and can process observations as they become available over time. The nonlinear reference trajectory is propagated after every observation, while the measurement correction remains in a separate state-deviation estimate. Because the sequential algorithm processes each observation once and does not globally relinearize the entire data arc, it can be more sensitive to a poor initial state or strong nonlinearity. Applications include autonomous vehicle and spacecraft navigation, GPS/INS systems, and time-critical or resource-limited estimation problems.

The batch filter processes all observations simultaneously, repeats nonlinear iterations to reduce linearization error, and generally provides a strong offline fit when the complete data set is available. However, it requires storing the observations, cannot update the estimate as soon as each observation arrives, and requires repeated full-arc trajectory and STM propagation. Applications include statistical orbit determination, offline trajectory reconstruction, and other non-real-time estimation environments where the complete data set is accessible.

5 Conclusion

This algorithm for orbit determination and propagation allowed for the development of valuable physical intuition about orbital mechanics. The batch filter results, computed using iterative least-squares estimation while accounting for atmospheric drag and Earth’s oblateness, showed that the filter converges after three iterations, reducing range residuals from approximately 732 m to approximately 0.01 m. These findings align closely with the theoretical expectations from the project description, which specifies measurement-noise levels of 0.01 m for range and 0.001 m/s for range-rate.

The range-only and range-rate-only comparison demonstrated that similar residual levels do not imply equivalent state information: the range-rate-only position covariance trace was approximately 7,400 times larger than the range-only result. The sequential implementation further exposed finite-precision covariance problems associated with the conventional covariance update and the highly scaled estimation state. Testing and adopting the Joseph form substantially improved the numerical covariance histories.

Ultimately, this project reinforced the importance of precision in computational methods and systematic validation through reliable input data, sensitivity modeling, covariance analysis, and iterative refinement, all of which are essential in orbit determination and propagation.

References

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